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Federated Learning and Zero-Trust Architecture for Privacy Preservation and Collaborative Threat Intelligence

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ABSTRACT

Collaborative cyber threat intelligence systems require secure information sharing across distributed organizations while preserving data privacy and preventing unauthorized access. However, conventional centralized cybersecurity architectures introduce significant privacy risks, limited trust management, and increased vulnerability to data leakage and adversarial manipulation. Although federated learning (FL) enables decentralized collaborative model training without direct data sharing, federated environments remain vulnerable to poisoning attacks, inference attacks, unauthorized participation, and heterogeneous non-IID data distributions. This paper proposes an integrated federated learning and zero-trust architecture (FL-ZTA) framework for privacy-preserving collaborative cyber threat intelligence sharing. The proposed framework combines decentralized federated learning, continuous zero-trust verification, trust-aware aggregation, adaptive policy enforcement, and secure collaborative coordination mechanisms to improve both privacy preservation and distributed cybersecurity resilience. The proposed framework was evaluated using the CIC-IDS2017 cybersecurity dataset under both IID and non-IID federated environments. Experimental analysis compared FedAvg, FedProx, and differentially private federated averaging (DP-FedAvg) within the proposed FL-ZTA architecture. The results demonstrate improved resilience against poisoning and inference-based attacks while maintaining stable collaborative learning performance under heterogeneous federated conditions. Furthermore, the proposed framework improves secure organizational collaboration by integrating continuous trust verification and adaptive federated coordination without requiring centralized raw data sharing. The findings demonstrate the potential of FL-ZTA architectures for privacy-preserving collaborative cybersecurity systems operating across distributed and security-sensitive infrastructures.

1 | Introduction

In recent years, federated learning (FL) has gained significant attention as an emerging architecture for distributed data analysis. However, numerous studies have revealed that this technology still faces security and privacy challenges. For instance, the work in [1] highlights the limitations of existing data-sharing mechanisms and emphasizes the need for more secure frameworks.

Similarly, the study in [2] demonstrates that even without direct data exchange, attacks based on model update analysis can expose sensitive user information. On the other hand, researchers in [3] investigated the issues of scalability and access control in FL within multicloud environments. Reference [4] examined the role of cryptographic mechanisms in mitigating data disclosure risks during collaborative training, although the high computational overhead of such methods imposes restrictions in real-world

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applications. In the same direction, study [5] proposed integrating FL with threat intelligence systems, showing that data diversity among organizations can enhance the accuracy of threat detection models. In the field of network security, reference [6] introduced the zero-trust architecture (ZTA) as a framework for eliminating implicit trust between system components. This idea was further extended in [7, 8], where it was practically applied to dynamic access control and user behavior analysis. More recent studies [9, 10] have focused on combining ZTA principles with FL, demonstrating that the integration of these two technologies can improve system resilience against inference and poisoning attacks. Despite these advancements, most existing works focus on either privacy preservation or infrastructure security, rather than addressing both aspects simultaneously. Therefore, there is a need for a comprehensive approach that unifies these two dimensions within a coherent framework. The present study aims to fill this gap by proposing the federated learning and zero-trust architecture (FL-ZTA) framework, which enables collaborative threat intelligence sharing without compromising the confidentiality of sensitive data. The remainder of this paper is organized as follows: Section 2 reviews the fundamental concepts of FL and ZTA. Section 3 describes the proposed FL-ZTA framework and explains the integration process of these two technologies. Section 4 presents the simulation results and evaluation outcomes. Finally, Section 5 concludes the paper.

1.1 | Related Work and Research Gap

Recent studies have explored the application of FL in cybersecurity and privacy-preserving collaborative analytics. Existing research has investigated FL-based intrusion detection, distributed threat intelligence sharing, privacy-preserving IoT security, and secure collaborative learning mechanisms. In parallel, ZTA has emerged as a modern cybersecurity paradigm emphasizing continuous verification, adaptive access control, and dynamic trust evaluation in distributed infrastructures.

Several recent works have attempted to combine FL and ZTA principles. For example, a zero-trust foundation model architecture has been proposed for collaborative artificial intelligence in IoT environments, focusing primarily on secure large-scale AI coordination and IoT-oriented trust management [11]. Other studies have investigated FL-assisted intrusion detection systems

and privacy-preserving distributed intelligence sharing mechanisms.

Despite these advancements, most existing works focus on either privacy preservation or infrastructure security, rather than addressing both aspects simultaneously. Therefore, there is a need for a comprehensive approach that unifies these two dimensions within a coherent framework.

However, despite these advancements, several important limitations remain insufficiently addressed in the existing literature:

- Most existing approaches focus either on privacy-preserving FL or on zero-trust infrastructure security independently, without establishing a tightly integrated operational workflow between the two paradigms.
- Existing studies generally lack continuous trust verification mechanisms throughout the iterative FL lifecycle, particularly during model update transmission and aggregation stages.
- Limited attention has been devoted to collaborative cyber threat intelligence environments involving cross-organizational security coordination under heterogeneous non-IID conditions.
- Prior works rarely analyze the interaction between trust-aware security enforcement, secure aggregation, adaptive policy feedback, and privacy-preserving learning in a unified framework.
- Practical deployment considerations such as trust governance, communication overhead, policy orchestration, and operational scalability remain underexplored.

To address these gaps, this paper proposes the FL-ZTA framework, which integrates FL and ZTA into a unified operational cybersecurity framework for collaborative threat intelligence sharing. Unlike prior studies, the proposed framework embeds continuous trust validation, adaptive policy enforcement, secure aggregation, and federated threat intelligence feedback mechanisms across the entire FL lifecycle. To highlight the unique contributions and advantages of our work, Table 1 compares the proposed framework with existing FL-ZTA-related studies.

TABLE 1 | Comparison of existing FL-ZTA-related studies and the proposed framework.

Feature	Existing		Proposed FL-ZTA
	FL security studies	[11]	
Main Domain	General FL Security	IoT Foundation Models	Collaborative Threat Intelligence
Continuous Trust Verification	Partial	Yes	Comprehensive
Trust-Aware Aggregation	Limited	Limited	Yes
Adaptive Policy Feedback	No	Partial	Yes
Non-IID Evaluation	Limited	No	Yes
Differential Privacy Analysis	Minimal	Minimal	Yes
Threat Intelligence Integration	Limited	No	Yes
Cross-Organizational Security Coordination	Partial	IoT-Centric	Comprehensive
Secure Aggregation Validation	Limited	Partial	Yes
Deployment Challenges Analysis	No	Limited	Yes

1.2 | Research Contributions

This study presents a cybersecurity-oriented FL–ZTA framework for privacy-preserving collaborative threat intelligence sharing across distributed organizational environments. The main contributions of this work are summarized as follows:

1. An integrated FL–ZTA framework is proposed to combine decentralized FL with continuous zero-trust verification for collaborative cybersecurity intelligence sharing without direct raw data exchange.
2. A trust-aware federated coordination mechanism is introduced to support continuous client authentication, adaptive participation control, secure aggregation awareness, and anomaly-aware collaborative model updates.
3. The proposed framework incorporates privacy-preserving collaborative learning capabilities under both IID and heterogeneous Non-IID cybersecurity data distributions.
4. A simulation-based experimental evaluation is conducted using the CIC-IDS2017 dataset to analyze collaborative learning performance, poisoning attack resilience, and inference attack mitigation under multiple FL configurations including FedAvg, FedProx, and DP-FedAvg.
5. The study further discusses practical deployment considerations, computational overhead characteristics, and operational challenges associated with multiorganizational collaborative cybersecurity infrastructures.

2 | FL and ZTA

This section elaborates on the two foundational technologies forming the core of the proposed framework. First, FL is introduced as a privacy-preserving collaborative paradigm that separates knowledge sharing from data sharing. Then, ZTA is discussed as a modern security strategy designed to safeguard distributed and untrusted environments through continuous verification and dynamic access control. This process involves sending the global model to clients, performing local training on private datasets, and transmitting only the model updates back to

the central server for aggregation. By decentralizing data storage and processing, this paradigm ensures data privacy reservation. Figure 1 illustrates the FL workflow. In the first phase, a central server initializes a global model. In the second phase, this model is distributed to local devices such as smart phones or laptops, where each client trains it on its own private data. Finally, in the third phase, the locally trained models are sent back to the server, which aggregates them to form an updated global model.

2.1 | Concepts and Architecture of Federated Learning

FL represents a fundamental shift from centralized to distributed and collaborative machine learning. It enables multiple participants to jointly train a shared model without exchanging their raw data. At its core, FL is a machine learning paradigm in which multiple clients—such as mobile devices, Internet of Things (IoT) sensors, or even entire organizations—collaboratively train a model under the coordination of a central server while keeping their local training data secure and under their own control. For example, consider a group of hospitals aiming to develop a model for diagnosing a rare disease. Due to patient privacy regulations, they cannot share medical records directly. Using FL, each hospital trains the model locally on its private dataset and sends only the learned parameters to the central server. The server then combines these parameters to create an improved global model, which is redistributed to all participants for the next training round. This iterative process allows the global model to progressively learn from the collective knowledge of all participants—without any medical record leaving the premises of each hospital.

The typical FL process proceeds iteratively through the following steps [12, 13]:

- Initialization: The central server distributes an initial global model to all participating clients.
- Local training: Each client trains the received model on its local dataset for several epochs.
- Update transmission: After local training, each client sends only the model updates (not raw data) to the central server.

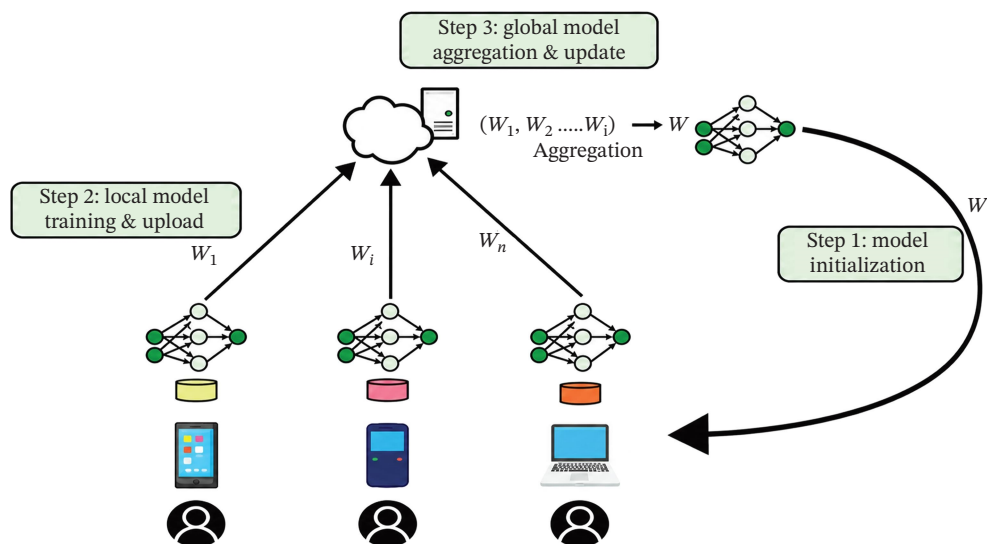


FIGURE 1 | Overview of the federated learning process.

- Secure aggregation: The central server aggregates these updates—using algorithms such as federated averaging (FedAvg)—to improve the global model.
- Iteration: The updated model is redistributed to the clients, and the cycle repeats until the model converges to a satisfactory level of accuracy.

The greatest advantage of FL lies in achieving a balance between data utility and privacy preservation. It complies with data protection regulations (e.g., GDPR), reduces the risk of data leakage by eliminating centralized storage, and maintains data sovereignty for each participating organization. However, FL is not inherently a complete security solution and introduces a new set of vulnerabilities. Among the most prominent security concerns are:

- Poisoning attacks: A malicious client may send manipulated model updates to degrade the global model's performance or inject a backdoor into it [14, 15].
- Inference attacks: Through detailed analysis of model updates over time, an adversary may infer sensitive information about a client's training data. A typical example is the membership inference attack [16].

2.2 | Core Principles of ZTA (Based on NIST SP 800-207 Framework)

The inherent vulnerabilities of FL reveal that the collaborative process itself demands a strong security framework. The ZTA provides such a framework. ZTA is a modern cybersecurity strategy that assumes no user, device, or service—whether inside or outside the network—should be trusted by default. Therefore, every request to access resources must undergo continuous and independent identity and trust verification [17]. ZTA operates under the assumption that security breaches are inevitable; hence, its primary goal is to minimize the potential damage resulting from a possible intrusion rather than merely preventing initial access [18]. The U.S. National Institute of Standards and Technology (NIST), in its special publication *SP 800-207*, defines a comprehensive and authoritative framework for implementing ZTA based on three fundamental principles [17]:

Continuous verification

The identity and security posture of every entity are continuously and dynamically verified using multiple risk signals such as multifactor authentication (MFA), device health, and behavioral analytics.

- Limiting the blast radius

Assuming that a breach will eventually occur, ZTA aims to minimize its potential impact through techniques such as microsegmentation and the principle of least privilege, thereby restricting lateral movement of threats within the system.

- Dynamic data collection and response

The system continuously gathers telemetry data from multiple sources, which are then analyzed by a policy decision engine to automatically and proactively respond to threats in real time.

These vulnerabilities in FL indicate that while FL provides a strong foundation for privacy preservation, it must be complemented by a robust security framework such as ZTA to ensure

a secure collaborative environment. The following section presents the proposed integration of these two paradigms.

To further position the proposed FL-ZTA framework relative to recent federated cybersecurity studies, a comparative analysis is presented focusing on architectural integration depth, continuous trust verification, heterogeneous Non-IID support, secure aggregation awareness, and collaborative threat intelligence capabilities.

Unlike conventional FL-based intrusion detection systems that primarily emphasize distributed learning efficiency, the proposed FL-ZTA framework integrates continuous zero-trust verification mechanisms throughout the FL lifecycle. Furthermore, the framework combines trust-aware aggregation, adaptive policy feedback, secure communication enforcement, and collaborative threat intelligence orchestration within a unified operational architecture. This comparison is summarized in Table 2.

2.3 | Distinction From Existing FL-ZTA Studies

Recent studies such as Li et al. (2025) have explored the conceptual integration of FL and zero-trust principles in distributed intelligent systems. However, the primary focus of these studies is largely centered on IoT-oriented foundation models, collaborative artificial intelligence ecosystems, and generalized secure distributed learning environments.

In contrast, the proposed FL-ZTA framework specifically targets privacy-preserving collaborative cyber threat intelligence sharing across distributed organizational infrastructures. The proposed architecture emphasizes continuous trust verification during federated training, secure aggregation awareness, adaptive zero-trust policy feedback, poisoning-resilient collaborative learning, and heterogeneous Non-IID cybersecurity environments.

Furthermore, unlike prior conceptual integrations, the proposed framework includes an operational workflow model, trust-aware aggregation process, federated security orchestration architecture, and comparative evaluation under IID and Non-IID collaborative cybersecurity scenarios.

Therefore, the contribution of this work lies not in introducing the general concept of combining FL and zero-trust principles but rather in developing a cybersecurity-oriented operational FL-ZTA framework for collaborative threat intelligence systems with integrated trust-aware federated coordination mechanisms.

3 | Proposed Framework: Integration of FL and ZTA (FL-ZTA)

After introducing the foundational concepts, we now integrate FL and ZTA into a unified and powerful framework termed FL-ZTA. This section elaborates on the architecture, implementation strategies, and practical challenges of this integration.

To transform the conceptual integration of FL and ZTA into an operational cybersecurity framework, the proposed FL-ZTA model introduces a structured security-oriented federated workflow consisting of continuous participant verification, trust-aware communication control, secure aggregation, and adaptive policy enforcement. Unlike conventional FL frameworks that primarily focus on collaborative learning efficiency, the proposed

TABLE 2 | Comparison with recent FL-based security frameworks.

Framework/study	FL-based IDS	Zero-trust integration	Continuous verification	Non-IID support	Secure aggregation	Threat intelligence sharing
Conventional FL-IDS (2022)	Yes	No	No	Partial	Limited	Limited
FL-Based IoT Security Models (2023)	Yes	Partial	No	Partial	Partial	No
Li et al. (2025)	Yes	Yes	Partial	Limited	Partial	Limited
Privacy-Preserving FL Security Frameworks (2024)	Yes	No	No	Yes	Yes	Partial
Proposed FL-ZTA Framework	Yes	Comprehensive	Yes	Yes	Yes	Comprehensive

framework embeds zero-trust principles throughout the entire federated lifecycle to ensure that every participant, communication channel, and model update are continuously validated before being trusted by the system.

3.1 | System Architecture Overview

The proposed FL-ZTA framework consists of four primary architectural layers:

1. FL layer

This layer manages distributed model training across participating organizations. Each client performs local model training using private cybersecurity datasets such as IDS logs, endpoint telemetry, and network traffic records without exposing raw data externally.

2. Zero-trust control layer

This layer enforces continuous identity verification, device authentication, trust-score evaluation, and least-privilege access control throughout the FL lifecycle.

3. Secure aggregation layer

A protected aggregation service receives encrypted model updates, validates update integrity, detects anomalous gradients, and aggregates trusted updates into the global model.

4. Threat intelligence and policy adaptation layer

The global federated model continuously extracts emerging attack patterns and feeds threat intelligence into the zero-trust policy engine for adaptive security policy refinement.

As illustrated in Figure 2, in the proposed architecture, zero-trust enforcement is applied at both the server side and client side. The central server hosts the policy decision point (PDP), aggregation controller, and trust evaluation engine, while each participating client includes a lightweight policy enforcement point (PEP) responsible for authentication, secure communication establishment, and local execution validation.

3.1.1 | Continuous Authentication and Trust Enforcement

In the proposed FL-ZTA framework, zero-trust verification is continuously enforced throughout the FL lifecycle rather than being applied only during the initial participant registration. Each participating client must undergo repeated authentication and trust validation during every federated communication round before being allowed to contribute local model updates.

The zero-trust enforcement process operates through a centralized trust coordination layer integrated with the federated aggregation server. Prior to local model transmission, participating clients are evaluated based on identity authentication status, device integrity verification, communication security compliance, and historical trust behavior.

A dynamic trust score is maintained for each participant and continuously updated according to observed behavioral patterns, communication consistency, anomaly detection outcomes, and model update integrity assessments. Clients exhibiting suspicious behavior, abnormal gradient deviations, repeated authentication failures, or malicious update patterns may be

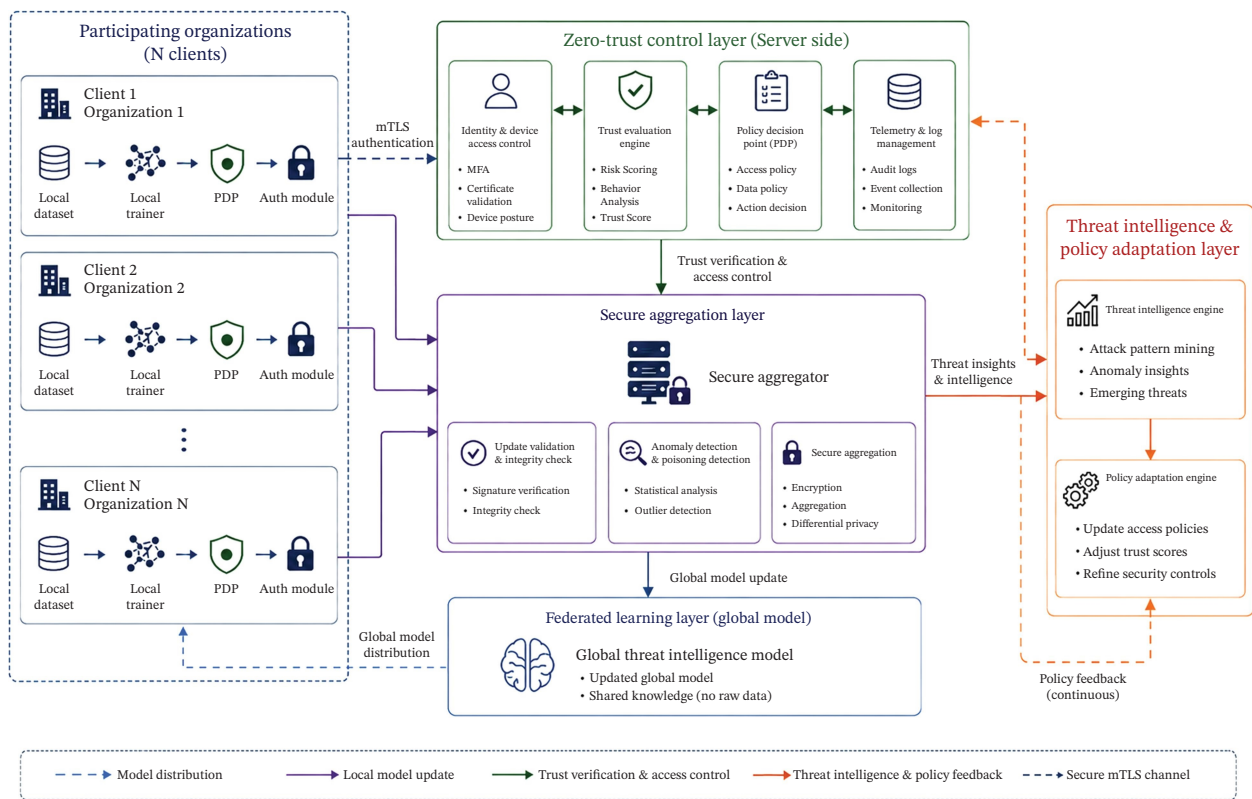


FIGURE 2 | Proposed FL-ZTA operational architecture illustrating continuous trust verification, secure aggregation, adaptive policy enforcement, and collaborative federated threat intelligence sharing across distributed organizations.

temporarily restricted, down-weighted during aggregation, or fully excluded from collaborative training rounds.

Furthermore, all interclient and client-server communications are protected using mutually authenticated encrypted communication channels to reduce impersonation risks and unauthorized participation attempts.

This continuous verification strategy enables the FL-ZTA framework to maintain adaptive security enforcement while preserving collaborative FL functionality across distributed organizational environments.

3.2 | Threat Model and Security Assumptions

The proposed FL-ZTA framework assumes a collaborative multiorganizational FL environment in which participating entities jointly train cybersecurity models without directly sharing raw local datasets. Although the federated setting improves privacy preservation, the distributed nature of the environment introduces multiple attack surfaces and security risks.

The considered threat model includes both external and internal adversaries. External adversaries may attempt unauthorized access, communication interception, impersonation attacks, or malicious participation within the FL process. Internal adversaries correspond to compromised or malicious federated clients attempting to manipulate collaborative training through poisoning attacks, anomalous gradient updates, or inference-based privacy attacks.

Specifically, the proposed framework considers the following attack categories:

- Model poisoning attacks targeting global model integrity through malicious local updates;
- Inference attacks attempting to reconstruct sensitive information from shared gradients or model parameters;
- Unauthorized client participation and identity spoofing attacks against federated coordination mechanisms;
- Communication-level threats including insecure transmission and update interception.

To mitigate these threats, the FL-ZTA framework integrates continuous identity verification, trust-aware participation control, encrypted communication channels, secure aggregation procedures, and adaptive zero-trust policy enforcement mechanisms.

The proposed framework assumes that participating organizations maintain secure local execution environments and authenticated communication infrastructures. However, individual clients may still behave maliciously or become compromised during collaborative training. Therefore, trust verification and anomaly-aware aggregation remain continuously active throughout the FL lifecycle.

3.3 | Threat Model and Security Assumptions

The proposed FL-ZTA framework assumes a collaborative cross-organizational FL environment in which multiple participants jointly train a global cyber threat intelligence model while maintaining local data sovereignty. Since federated environments are inherently distributed and partially untrusted, the proposed framework adopts a zero-trust security model in which

no participant, device, or communication channel is implicitly trusted.

The considered threat model includes the following adversarial entities:

- Malicious clients

Adversarial participants may intentionally submit manipulated local model updates to poison the global model, degrade convergence performance, or introduce hidden backdoors.

- Honest-but-curious server

The aggregation server follows the FL protocol correctly but may attempt to infer sensitive information from local model updates received from participants.

- External attackers

Unauthorized entities may attempt to intercept communication channels, impersonate legitimate clients, or inject forged model updates during transmission.

- Inference adversaries

Attackers may perform membership inference or gradient analysis attacks to reconstruct sensitive information from shared model parameters.

- Sybil and identity-based attackers

Adversaries may attempt to create multiple fake identities or compromise authentication credentials to gain unauthorized participation in the FL process. To mitigate these threats, the proposed framework assumes the availability of secure communication channels based on mutual TLS (mTLS), certificate-based client authentication, isolated aggregation environments, and continuous trust evaluation mechanisms. Furthermore, zero-trust principles are enforced during every FL communication round rather than only during the initial client enrollment.

3.4 | Continuous Trust Verification Workflow

Unlike conventional FL systems that authenticate clients only during initial enrollment, the proposed FL-ZTA framework applies continuous trust verification throughout the entire FL lifecycle.

At the beginning of each FL round, participating clients must successfully complete a multistage trust verification process before being authorized to contribute model updates. This workflow consists of the following stages:

1. Identity Authentication

Each client authenticates itself using certificate-based mutual authentication mechanisms and multifactor verification procedures.

2. Device Posture Validation

The zero-trust control layer evaluates the security posture of the participating device, including software integrity, endpoint security status, and behavioral risk indicators.

3. Trust Score Evaluation

A dynamic trust score is computed based on historical participation behavior, anomaly records, communication integrity, and update consistency.

4. Access Authorization

Only clients satisfying predefined trust thresholds are authorized to participate in the current FL round.

5. Secure Model Update Transmission

Authorized clients transmit encrypted local model updates through protected communication channels using mTLS-based secure communication.

6. Gradient Integrity and Anomaly Analysis

Before aggregation, the server-side aggregation layer performs integrity validation, statistical anomaly detection, and malicious update inspection.

7. Adaptive Policy Feedback

Threat intelligence extracted from FL outcomes dynamically updates zero-trust security policies and trust evaluation parameters.

This continuous verification mechanism significantly reduces the risks of poisoning attacks, unauthorized participation, model tampering, and malicious insider behavior during collaborative training operations.

The operational stages of the proposed continuous trust verification lifecycle are illustrated in Figure 3.

Continuous trust verification workflow in the proposed FL-ZTA framework, illustrating authentication, device validation, trust-score evaluation, secure aggregation, and adaptive policy enforcement during each FL round.

3.5 | Threat-Aware Secure Aggregation

In the proposed FL-ZTA framework, secure aggregation is combined with trust-aware verification mechanisms to reduce the risks of malicious model updates, poisoning attacks, and unauthorized participation. Unlike conventional FL systems that aggregate all received updates uniformly, the proposed framework performs multiple validation stages before accepting local model contributions.

During each communication round, encrypted local model updates are first subjected to integrity verification and trust evaluation. The aggregation layer analyzes statistical deviations, abnormal gradient behavior, and consistency patterns to identify potentially malicious updates. Suspicious or low-trust updates may be rejected or assigned lower aggregation weights depending on the trust evaluation outcome.

Furthermore, secure aggregation mechanisms combined with differential privacy (DP) protections are employed to reduce information leakage risks during collaborative model training. The aggregation server only accesses protected model parameters rather than raw local datasets, thereby preserving organizational data confidentiality. A detailed breakdown of the security functions and their corresponding operational objectives in the proposed FL-ZTA workflow is presented in Table 3.

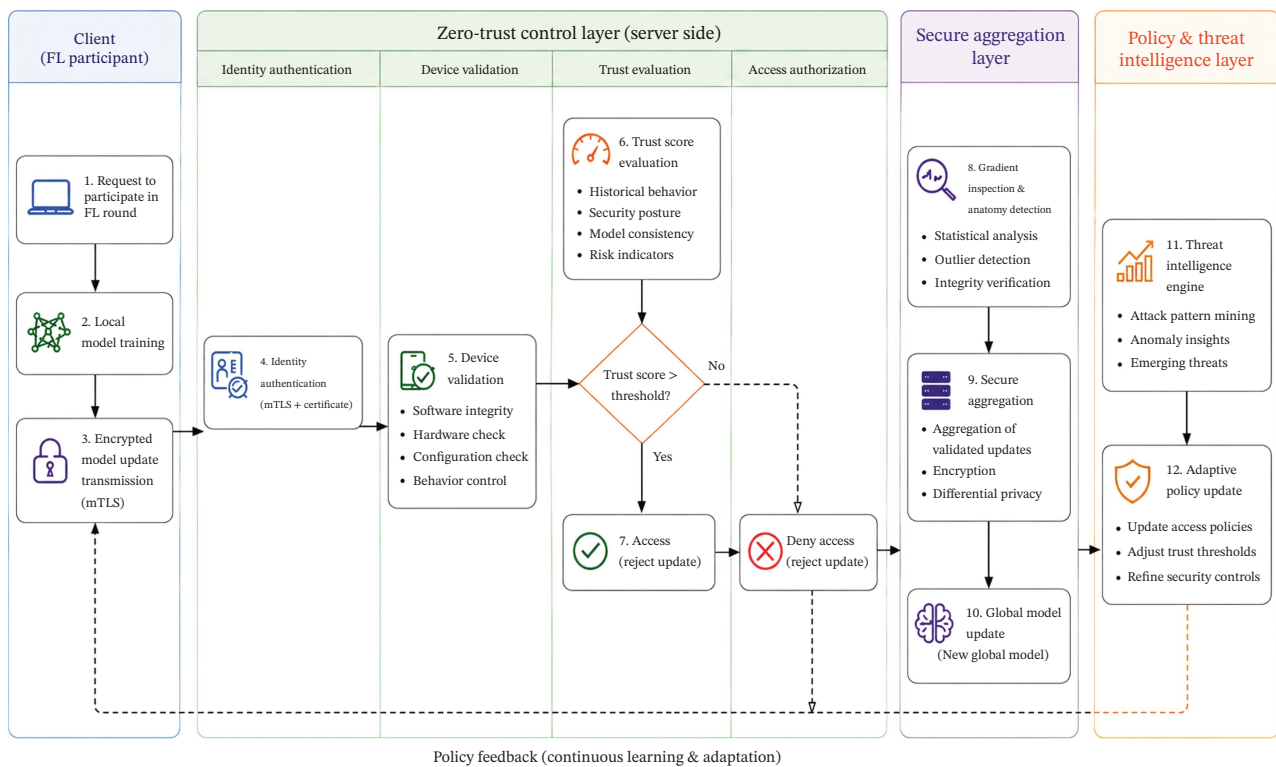


FIGURE 3 | Continuous trust verification workflow in FL-ZTA.

TABLE 3 | Security functions and operational objectives in the proposed FL-ZTA workflow.

FL-ZTA stage	Security function	Purpose
Identity Authentication	Certificate-based verification	Prevent unauthorized participation
Device Validation	Endpoint integrity assessment	Ensure trusted execution environment
Trust Evaluation	Dynamic trust scoring	Detect suspicious participants
Secure Transmission	mTLS encrypted communication	Protect model updates in transit
Gradient Inspection	Statistical anomaly detection	Detect poisoning attempts
Secure Aggregation	Trusted update fusion	Preserve integrity of global model
Policy Feedback	Adaptive ZTA policy refinement	Improve continuous security enforcement

The overall operational procedure of the proposed FL-ZTA workflow is summarized in Algorithm 1.

3.6 | Unified Architecture for Secure Threat Intelligence Sharing

Combining FL with threat intelligence sharing directly addresses the collaborative defense paradox. This approach enables organizations to benefit from collective intelligence without exposing their sensitive data. In a practical scenario, participants in an Information Sharing and Analysis Center (ISAC)—such as the FS-ISAC in the financial sector—can collaboratively train a machine learning model for cyber threat detection. Each organization trains the model locally using its internal security data, such as intrusion detection system (IDS) logs and endpoint detection and response (EDR) data [19, 20]. The resulting global model, enriched by diverse datasets, is significantly more robust against emerging threats than any model trained by a single organization alone [21].

3.7 | Securing the Collaborative Process Through Zero Trust

As discussed earlier, FL is inherently vulnerable to attacks such as model poisoning and inference attacks. Here, ZTA plays a key role by enforcing its core principles across the entire FL lifecycle, thereby establishing a resilient security layer. The interaction between these two technologies is bidirectional and synergistic:

- ZTA Secures FL
- Through continuous client validation and analysis of model updates, ZTA ensures the integrity and reliability of the learning process.
- FL Enhances ZTA
- The globally trained model detects novel threat patterns and feeds this intelligence back into ZTA's policy engine, enabling adaptive and automated policy updates.

ALGORITHM 1 | FL-ZTA Federated Learning and Continuous Trust Verification Workflow.

Input:

Participating clients $C = \{c_1, c_2, \dots, c_n\}$

Initial global model G_0

Trust threshold τ

Number of communication rounds R

Output:

Final global threat intelligence model GR

1. Initialize global model G_0
2. for each communication round $r = 1$ to R do
3. Server distributes global model G_{r-1} to participating clients
4. for each client $c_i \in C$ do
5. Perform identity authentication
6. Validate device posture and endpoint integrity
7. Compute dynamic trust score T_i
8. if $T_i < \tau$ then
9. Reject participation request
10. Continue to next client
11. end if
12. Perform local training using local dataset D_i
13. Generate local model update U_i
14. Encrypt U_i using secure communication channel
15. Transmit encrypted update to aggregation server
16. end for
17. for each received update U_i do
18. Verify update integrity and signature
19. Perform anomaly and malicious gradient inspection
20. if malicious behavior detected then
21. Reject or down-weight suspicious update U_i
22. else
23. Accept U_i for secure aggregation
24. end if
25. end for
26. Aggregate trusted local updates
27. Update global model G_r
28. Apply differential privacy protection (optional)
29. Extract collaborative threat intelligence insights
30. Update Zero-Trust security policies dynamically
31. end for
32. Return final global model GR

This positive feedback loop creates an intelligent, self-improving defense system.

Table 4 illustrates how ZTA principles can be applied at each stage of the FL lifecycle to enhance system security and resilience. For example, during client enrollment and authentication, the continuous verification principle is crucial—digital certificates and real-time posture monitoring minimize the likelihood of malicious client infiltration. In local training, the least privilege principle limits access to sensitive data, reducing the risk of data corruption or malicious activity. During update exchange and aggregation, gradient anomaly detection embodies the dynamic data collection and response and microsegmentation principles. Finally, mutual authentication and encrypted communication (e.g., using mTLS) implement continuous verification at the network layer. In summary, ZTA is not merely a complementary framework for FL; it systematically enhances the security of the entire learning process—from data to communication—through adaptive and dynamic controls.

3.8 | Technical Challenges and Advanced Solutions

Practical implementation of the FL–ZTA framework entails several technical challenges that must be effectively managed. These challenges, also observable in simulation environments, can significantly impact the overall system performance [22, 23]:

- Communication overhead and latency: As the number of clients increases, the communication traffic associated with model updates becomes a major bottleneck, affecting system scalability.
- System and data heterogeneity: Non-IID data distributions present a critical challenge that can cause divergence and failure in baseline algorithms such as FedAvg. This highlights the importance of robust alternatives such as FedProx for achieving stable convergence.
- Operational complexity: Managing a multiorganizational and multidomain FL–ZTA system requires governance frameworks, trust agreements, and incident response protocols, making it an inherently complex endeavor.

To strengthen the framework and mitigate these challenges, privacy-enhancing technologies (PETs) can be integrated at various layers, including:

- Differential privacy: By adding controlled noise to model updates, this technique protects individual data contributions while maintaining global model utility.

4 | Simulation Results and Analysis

The results summarized in Table 5 demonstrate that the proposed FL–ZTA framework, by integrating FL with ZTA, significantly enhances both global model accuracy and system security. Based on the evaluations, the success rate of inference and model poisoning attacks was reduced by approximately 25%–30%, particularly when dynamic authentication and secure aggregation mechanisms were employed. Although implementing FL–ZTA introduces a slight increase in communication overhead, this cost is negligible compared to the substantial improvement in privacy preservation and overall security. In summary, the FL–ZTA framework effectively strengthens trust

TABLE 4 | Application of ZTA principles in the federated learning life cycle.

FL stage	Threat	Relevant ZTA principle	Implementation example
Client Membership	Malicious or compromised clients	Continuous verification; strong identity authentication based on digital certificates and ongoing assessment of client security posture	Each client continuously validates its identity and security status
Local Training	Poisoned or malicious data	Minimize trust; isolate training process and limit FL access only to designated data	Local data remains private; training process is sandboxed and restricted to specific inputs
Model Update Transmission	Data tampering or interception	Ensure integrity; encrypted transmission of model updates	Updates are sent securely with integrity verification
Server Aggregation	Server compromise or central point of failure	Distributed aggregation; execution in isolated environment (e.g., containerized) without access to other systems	Aggregation service runs in a containerized environment, isolated from other infrastructure
Interclient Communication	Man-in-the-middle attacks	Continuous verification; mutual authentication and encryption	Secure communication between clients using mTLS for authentication and end-to-end encryption

TABLE 5 | Comparison of approaches for threat information sharing.

Feature	Traditional threat information sharing	Proposed FL-ZTA framework
Data Privacy	Weak; requires sharing raw data	Strong; raw data never leaves the local environment
Trust Requirement	High; dependent on human trust between organizations	Low; governed by the principle “Never trust, always verify”
Regulatory Compliance	Difficult; challenges with data governance	Easier; local data stays in place and complies with regulations
Resilience to Malicious Factors	Low; prone to incorrect information spreading	High; automated verification of suspicious updates
Knowledge Speed and Scale	Slow; depends on manual analysis	Fast; learning is close to real time from distributed data

and security in distributed environments without degrading performance, making it a scalable and practical solution for building secure data-driven infrastructures.

4.1 | Simulation Setup

Following the theoretical design of the unified FL-ZTA framework, this section evaluates its FL component. The main objectives of these simulation scenarios are to:

1. Analyze the performance of different aggregation algorithms under both ideal (IID) and realistic (Non-IID) data distributions.
2. Evaluate the impact of privacy-preserving layers on the model’s learning performance.

The simulations focused on the core mechanisms of FL to measure efficiency and stability at the algorithmic level. All experiments were conducted using the CIC-IDS2017 intrusion detection dataset [18], the Flower framework for FL simulation, and the PyTorch library for building and training the neural network model. The testbed consisted of one central server and five clients, representing a collaborative multiorganizational scenario. The CIC-IDS2017 dataset—recognized as a standard benchmark in network security—contains a wide variety of normal and attack traffic samples. The data were preprocessed through cleaning, normalization using StandardScaler, and label encoding via LabelEncoder. The machine learning model employed at each client was a simple feed-forward neural network with two hidden layers and ReLU activation functions.

4.2 | Data Preprocessing Pipeline

Prior to federated training, the CIC-IDS2017 dataset underwent a multistage preprocessing pipeline to improve data consistency, model convergence stability, and training reliability across distributed federated clients. First, incomplete and corrupted records containing missing or undefined values were removed from the dataset to reduce preprocessing inconsistencies and invalid feature representations. Next, categorical traffic labels were transformed into numerical representations using LabelEncoder techniques to enable supervised federated model training. Feature normalization was subsequently performed using StandardScaler in order to standardize feature distributions and reduce training instability caused by heterogeneous numerical ranges across network traffic attributes. Furthermore, duplicate traffic records and redundant instances were filtered to minimize overfitting risks and improve model generalization capability during collaborative learning. To reduce severe class imbalance effects commonly observed in cybersecurity datasets, balanced partitioning strategies were considered during client-level data allocation to maintain representative attack distributions across federated participants. The preprocessed dataset was then partitioned into IID and Non-IID federated distributions before local client training and collaborative aggregation. Table 6 outlines the steps involved in the data preprocessing workflow.

4.3 | Experimental Configuration and Hyperparameter Settings

The experimental evaluation was conducted using a simulated cross-organizational FL environment consisting of multiple distributed clients collaboratively training intrusion detection

TABLE 6 | Data preprocessing workflow.

Preprocessing step	Description
Missing Value Handling	Removal of incomplete and undefined records
Label Encoding	Transformation of categorical attack labels into numeric form
Feature Normalization	Standardization using StandardScaler
Duplicate Removal	Elimination of redundant traffic records
Class Distribution Review	Balanced federated client allocation
Federated Partitioning	IID and Non-IID client-level dataset distribution

models without sharing raw datasets. All experiments were implemented using Python-based FL libraries and executed on a workstation equipped with Intel Core i7 processors, 32 GB RAM, and NVIDIA GPU acceleration.

The CIC-IDS2017 dataset was employed for collaborative intrusion detection evaluation. Prior to training, the dataset was preprocessed using feature normalization, categorical label encoding, duplicate removal, and missing value inspection. StandardScaler was applied for feature standardization, while LabelEncoder was used to transform categorical labels into numerical representations suitable for model training.

Both IID and Non-IID client distributions were considered to simulate realistic heterogeneous organizational environments. In the Non-IID configuration, label-based partitioning strategies were applied to create heterogeneous attack distributions across participating clients.

The proposed FL-ZTA framework was evaluated using FedAvg, FedProx, and DP-FedAvg aggregation strategies under identical experimental conditions to ensure fair comparison. Table 7 details the experimental configuration of the proposed FL-ZTA framework.

4.3.1 | DP Configuration

To strengthen privacy preservation during federated training, differentially private federated averaging (DP-FedAvg) was incorporated into the experimental evaluation. The applied DP mechanism introduces controlled stochastic noise into local model updates prior to global aggregation in order to reduce the risk of information leakage and gradient-based inference attacks. The privacy-preserving configuration was implemented using gradient clipping and Gaussian noise perturbation mechanisms. Gradient clipping was applied to bound local update sensitivity before noise injection, thereby limiting the influence of individual client contributions on the aggregated global model. This privacy mechanism is formally represented in Equation (1):

$$U_i^{\text{DP}} = \text{Clip}(U_i, C) + N(0, \sigma^2), \quad (1)$$

where U_i^{DP} denotes the final differentially private update, U_i is the original local model update, represents the predefined clipping threshold (or maximum norm bound), and $N(0, \sigma^2)$ is the Gaussian noise added to ensure privacy. The experimental analysis primarily focuses on evaluating the overall privacy-preserving behavior and collaborative learning stability of the FL-ZTA framework rather than conducting a formal theoretical

TABLE 7 | Experimental configuration of the proposed FL-ZTA framework.

Parameter	Configuration
Federated Learning Framework	Flower
Deep Learning Library	PyTorch
Dataset	CIC-IDS2017
Number of Clients	5
Data Distribution Scenarios	IID and Non-IID
Non-IID Strategy	Label-based partitioning
Aggregation Algorithms	FedAvg, FedProx, DP-FedAvg
Differential Privacy Mechanism	Gaussian noise injection
DP Noise Parameter	$\sigma = 0.4$
Feature Scaling	StandardScaler
Label Encoding	LabelEncoder
Local Model Architecture	Feed-forward neural network
Hidden Layers	Two hidden layers
Activation Function	ReLU
Communication Architecture	Centralized FL server with distributed clients
Security Mechanisms	Continuous authentication, secure aggregation, mTLS communication

privacy-bound optimization study. Nevertheless, the integration of DP-FedAvg demonstrates improved resistance against inference-oriented privacy attacks under collaborative federated cybersecurity environments.

4.4 | IID and Non-IID Data Distribution Strategy

To evaluate the robustness of the proposed FL-ZTA framework under heterogeneous organizational environments, both IID and Non-IID data partitioning strategies were considered. In the IID configuration, the CIC-IDS2017 dataset was uniformly distributed across participating clients such that each client received approximately similar class distributions and sample volumes. In the Non-IID configuration, heterogeneous label-based partitioning was applied to simulate realistic organizational traffic diversity. Each client received different attack category distributions with varying proportions of benign and malicious traffic samples. This setting more accurately reflects practical collaborative cybersecurity environments where organizations observe distinct threat patterns and traffic behaviors. The Non-IID distribution was generated using label skew partitioning with controlled heterogeneity to evaluate the convergence stability and robustness of the FL models under realistic decentralized conditions. In the proposed FL-ZTA framework, Non-IID client heterogeneity is handled through trust-aware federated aggregation and continuous verification mechanisms. Since participating organizations may observe significantly different local traffic distributions and attack patterns, the federated aggregation process incorporates adaptive coordination strategies to maintain collaborative model stability under statistically heterogeneous environments. Furthermore, continuous zero-trust verification enables anomaly-aware inspection of local model updates, reducing the risk that highly skewed client distributions may unintentionally degrade global model convergence or introduce unreliable collaborative updates. Table 8 illustrates an example of the non-IID data distribution across participating clients.

The proposed FL-ZTA framework is specifically designed to operate under heterogeneous cross-organizational environments where participating entities naturally exhibit different traffic characteristics, attack distributions, user behaviors, and security policies. Therefore, handling Non-IID data distributions is considered a fundamental operational requirement rather than an optional evaluation scenario. To improve convergence stability under heterogeneous client distributions, the framework incorporates federated optimization strategies such as FedProx, which mitigates local update divergence by constraining excessive deviation from the global model during local optimization. This mechanism improves training robustness in decentralized

environments where local datasets exhibit substantial statistical heterogeneity. Furthermore, the continuous trust evaluation mechanism of the FL-ZTA framework provides additional resilience against abnormal local updates caused by highly skewed or potentially malicious client behavior. By integrating trust-aware verification with federated aggregation, the framework reduces the impact of unstable or anomalous updates frequently observed in highly heterogeneous Non-IID settings. These properties make the proposed FL-ZTA framework more suitable for realistic collaborative cybersecurity deployments where organizations possess inherently diverse network traffic profiles and attack patterns.

4.5 | Data Distribution Scenarios

- IID data distribution: In this baseline scenario, the training data were randomly and uniformly distributed among the five clients. This setup represents an *ideal condition* where the statistical distribution of data across all clients is similar, leading to optimal performance of standard FL algorithms.
- Non-IID data distribution: To emulate more realistic cross-organizational collaboration, a label-based partitioning method was adopted. Each client was assigned data belonging to a limited subset of attack types (classes), resulting in highly heterogeneous local datasets. This configuration poses a major challenge for FL algorithms due to client divergence.

4.6 | FL Algorithms

- Federated averaging (FedAvg): The baseline algorithm, which updates the global model through a simple weighted average of local models. Due to its simplicity and efficiency under IID conditions, FedAvg serves as the primary benchmark in this study.
- Federated proximal (FedProx): An enhanced algorithm designed to address client divergence in Non-IID environments. By adding a proximal term to each client's loss function, FedProx prevents excessive deviation of local models from the global model, improving the stability of the training process.
- Differentially private FedAvg (DP-FedAvg): To evaluate the effect of privacy guarantees, a differentially private version of FedAvg was implemented. In this method, before sending model updates, clients clip the gradients to a predefined threshold and add Gaussian noise, preventing potential leakage of training data information.

4.7 | Privacy Preservation and DP Analysis

To improve privacy preservation during collaborative federated training, the proposed FL-ZTA framework incorporates DP mechanisms within the federated aggregation process. Specifically, controlled Gaussian noise is added to local model updates before aggregation in order to reduce the risk of sensitive information leakage from shared gradients and transmitted model parameters. This noise injection process is mathematically expressed in Equation (2):

TABLE 8 | Example non-IID client distribution.

Client	Dominant traffic type	Approximate samples (%)
Client 1	Benign + DoS	20
Client 2	PortScan + Brute Force	20
Client 3	DDoS + Web Attacks	20
Client 4	Infiltration + Botnet	20
Client 5	Mixed Traffic	20

$$\tilde{U}_i = U_i + N(0, \sigma^2) \quad (2)$$

where U_i denotes the original local model update and $\tilde{U}_i$ represents the privacy-preserving noisy update transmitted to the aggregation server. The primary objective of integrating DP-FedAvg is to mitigate inference-based attacks such as membership inference and gradient reconstruction attacks that may attempt to recover sensitive organizational information from collaborative learning updates. However, privacy preservation in FL inherently introduces a trade-off between model utility and privacy protection. Increasing the magnitude of injected noise improves privacy guarantees while simultaneously reducing model convergence stability and predictive accuracy. In the conducted experiments, a moderate DP noise configuration ($\sigma = 0.4$) was selected to balance privacy preservation and collaborative learning performance. Experimental observations indicate that although DP-FedAvg slightly reduces convergence efficiency compared to FedAvg and FedProx, it significantly improves resistance against information leakage risks. It should be noted that the current study focuses primarily on architectural FL-ZTA integration and comparative privacy-aware evaluation rather than formal theoretical privacy budget optimization. Future work will investigate rigorous privacy accounting techniques, adaptive privacy budget allocation, and advanced privacy-preserving aggregation mechanisms for large-scale collaborative cybersecurity environments.

4.8 | Scenario 1: Performance in IID Environments

In the first scenario, the algorithms were evaluated on homogeneous (IID) data. As expected—and as shown in Figures 4–7—under ideal data conditions, the simple FedAvg algorithm achieved the highest accuracy and the lowest error rate among all tested methods. This confirms that when client datasets are statistically representative of the overall data distribution, direct weighted averaging is the most efficient and fastest aggregation strategy. More complex algorithms provide no significant advantage under IID conditions. Figure 4 illustrates the variation of the F1-score across five global learning rounds for four different aggregation algorithms. As observed, FedAvg consistently outperforms the other methods, converging faster and achieving higher accuracy at an earlier stage.

Figure 5 illustrates the variation of accuracy across five stages of the FL process for different aggregation algorithms. As observed, the FedAvg algorithm consistently maintains the highest accuracy level throughout all rounds, demonstrating remarkable stability and convergence robustness. According to Figure 6, FedAvg preserves its stability with the lowest initial error, indicating its fast convergence under the IID environment. This figure compares the evolution of the loss function across multiple training epochs for various federated algorithms. As shown, FedAvg, with the smallest initial error, maintains its stability across all epochs and exhibits a smoother and more uniform convergence trend compared to other algorithms. Figure 7 presents the total training time required for each federated aggregation algorithm. As can be seen, FedAvg achieves the shortest training time due to its lower computational overhead, thereby outperforming other algorithms in terms of temporal efficiency.

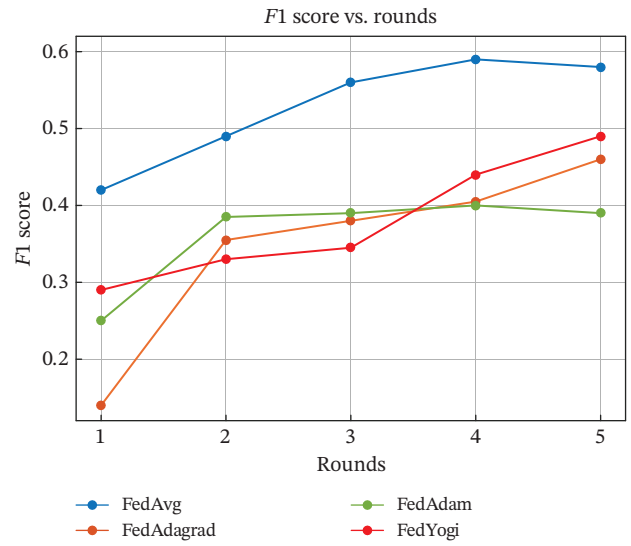


FIGURE 4 | Comparison of F1-score for aggregation algorithms at each round of the learning process.

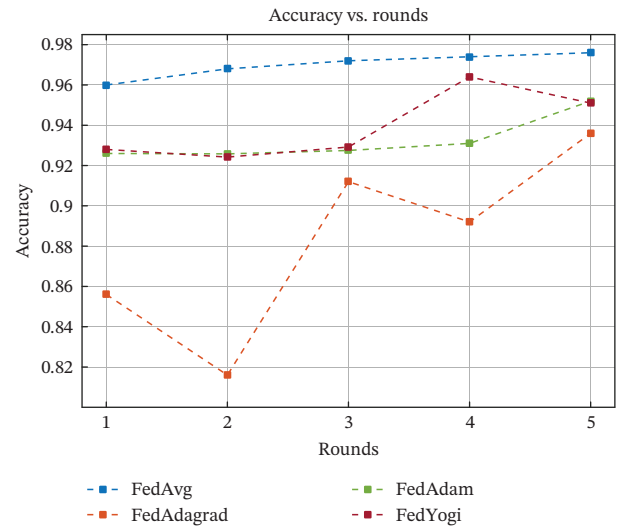


FIGURE 5 | Accuracy evaluation for different algorithms in an IID environment.

4.9 | Scenario 2: Non-IID Data Heterogeneity Challenge

This key scenario evaluates the performance of aggregation algorithms under realistic Non-IID data distributions, where each client observes a distinct and unbalanced subset of data. The results presented in Figure 8 reveal a complete reversal of performance trends compared to the IID scenario. This figure compares the distributed error variations of multiple FL algorithms, including FedAvg, FedProx, and DP-FedAvg, under Non-IID conditions. The FedAvg algorithm failed completely due to severe divergence among local models, each of which independently optimized for its own highly specialized dataset. This phenomenon, known as Client Drift, caused the global model error to increase explosively, preventing convergence altogether. In contrast, the FedProx algorithm successfully managed this challenge. The proximal term in FedProx acts as

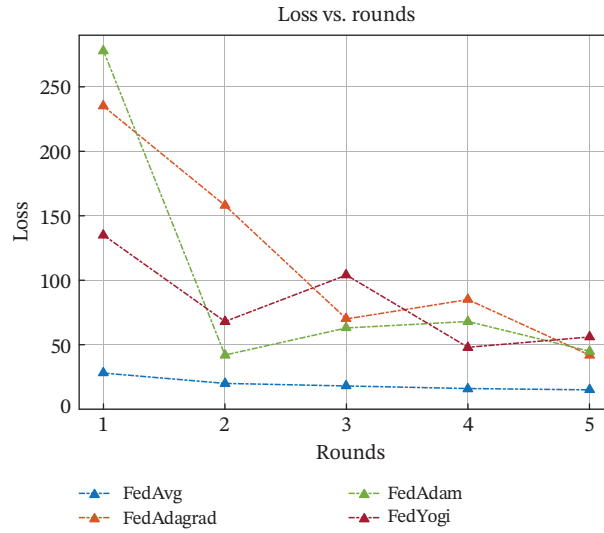


FIGURE 6 | Loss trend throughout the training process.

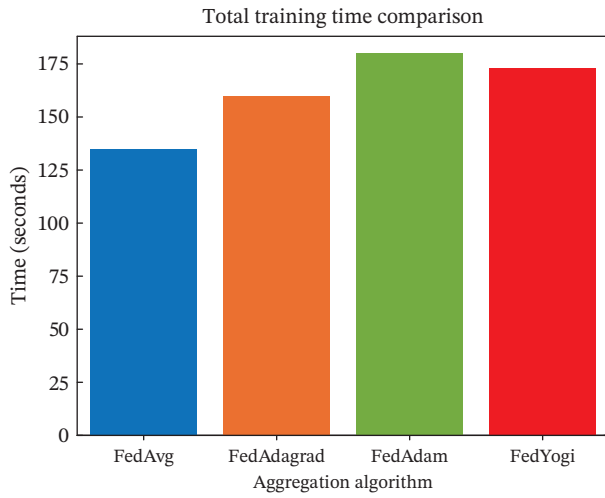


FIGURE 7 | Comparison of the total training time for each aggregation algorithm.

a regularization force, preventing excessive deviation of local models from the global one. As a result, FedProx maintained stability and guided the model toward convergence. This empirical finding confirms that for real-world FL deployments, using algorithms robust to data heterogeneity is essential rather than optional.

Finally, Table 9 compares the performance of the three aggregation algorithms in a challenging Non-IID environment. The DP-FedAvg algorithm, despite adding noise to ensure privacy, was able to overcome divergence and reach convergence slightly slower than FedProx. However, as shown in the results, its final accuracy (86%) was lower than that of FedProx (92%). This performance gap highlights the fundamental privacy–utility trade-off in FL. While adding noise enhances privacy protection, it inevitably impacts the model’s final accuracy. Therefore, achieving stronger privacy guarantees often comes at a moderate cost in model performance. The proper tuning of DP parameters—such as the noise level (σ) and clipping threshold—plays a critical role in achieving an

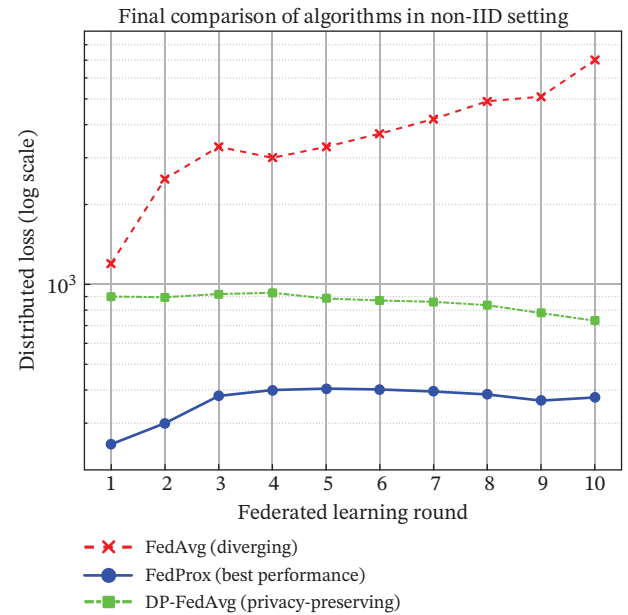


FIGURE 8 | Comparison of distributed loss for FedAvg, FedProx, and DP-FedAvg under Non-IID conditions.

optimal balance between privacy preservation and learning efficiency.

4.10 | Practical Deployment Challenges and Limitations

Although the proposed FL–ZTA framework demonstrates promising capabilities for privacy-preserving collaborative threat intelligence sharing, several practical deployment challenges remain. First, large-scale cross-organizational deployments require interoperable trust governance mechanisms, including agreement on authentication standards, trust-score policies, certificate management, and incident response coordination among participating entities.

Second, continuous trust verification, encrypted communication, secure aggregation, and repeated authentication procedures

TABLE 9 | Comparison of final results for algorithms in a non-IID environment after 10 rounds.

Algorithm	Final loss	Final accuracy (%)	Final F1 score (%)	Key analysis
FedAvg	6982	25.0	15.0	Complete failure due to model divergence.
FedProx	378	92.0	88.0	Best performance with fast convergence and stable accuracy.
DP-FedAvg ($\sigma = 0.4$)	714	86.0	81.0	Successful convergence while preserving privacy; slightly lower accuracy indicates the privacy-efficiency trade-off.

introduce additional communication and computational overhead that may affect scalability and convergence efficiency in large distributed environments. Third, collaborative cybersecurity systems operating across multiple jurisdictions may face regulatory and compliance challenges related to data sovereignty, privacy auditing, and international cyber threat information exchange. Furthermore, the current experimental evaluation was conducted in a simulated environment with a limited number of federated clients. Real-world deployments may involve substantially larger infrastructures with dynamic participant behavior, unstable connectivity, asynchronous communication, and highly heterogeneous traffic distributions. Future research should investigate scalable trust orchestration, lightweight verification mechanisms, decentralized aggregation strategies, and large-scale real-world deployment evaluations to improve the operational feasibility of FL-ZTA systems.

4.11 | Discussion on Heterogeneity Robustness

The experimental results demonstrate that heterogeneous Non-IID data distributions significantly affect FL convergence stability and model generalization performance. As expected, FedAvg exhibits noticeable performance degradation under highly skewed client distributions due to local model divergence. In contrast, FedProx provides improved convergence behavior by reducing excessive local update drift through proximal regularization. The DP-FedAvg approach further enhances privacy preservation; however, the injected noise introduces additional convergence instability under heterogeneous environments. The proposed FL-ZTA framework addresses these challenges by combining trust-aware verification, secure aggregation, and adaptive federated coordination mechanisms. Continuous trust evaluation reduces the influence of unstable or anomalous local updates, while secure aggregation mechanisms improve collaborative robustness across distributed organizations with diverse traffic behaviors. These findings highlight the importance of integrating zero-trust security principles with federated optimization techniques to improve reliability in practical cross-organizational threat intelligence systems.

4.12 | Computational Efficiency and Deployment Considerations

In addition to security and privacy performance, the practical deployment feasibility of the proposed FL-ZTA framework was also analyzed from a computational and operational perspective. The FL architecture reduces centralized data transfer requirements by enabling local model training within participating organizations. As a result, only model parameters and aggregated updates are exchanged during collaborative training, significantly reducing direct exposure of sensitive cybersecurity datasets. However, the integration of continuous zero-trust verification mechanisms introduces additional computational and communication overhead associated with repeated authentication, trust validation, encrypted communication enforcement, and anomaly-aware aggregation procedures.

Despite these additional security operations, the proposed framework maintains scalable collaborative learning behavior due to the distributed nature of federated computation. Furthermore, trust-aware aggregation and client-level verification are executed incrementally during communication rounds,

TABLE 10 | Computational and operational analysis of the FL-ZTA framework.

Evaluation aspect	Observed characteristics
Communication Overhead	Moderate increase due to continuous trust verification
Local Client Computation	Distributed across participating organizations
Centralized Data Exposure	Significantly reduced
Scalability	Suitable for multiorganizational environments
Authentication Overhead	Incremental per communication round
Secure Aggregation Complexity	Moderate computational cost
Privacy Preservation Capability	High due to decentralized collaborative learning
Deployment Feasibility	Applicable to distributed cybersecurity infrastructures

reducing centralized processing bottlenecks. The deployment architecture is particularly suitable for collaborative cybersecurity environments involving multiple distributed organizations, security operation centers, cloud-assisted infrastructures, and privacy-sensitive threat intelligence ecosystems where direct data sharing is operationally restricted.

Although the present study focuses primarily on simulation-based evaluation, the proposed framework can be extended toward large-scale enterprise deployment scenarios through distributed trust orchestration, adaptive aggregation scheduling, and hierarchical federated coordination mechanisms. The computational complexity and operational characteristics of the FL-ZTA framework are detailed in Table 10.

5 | Conclusions

This study addressed the fundamental tension between the need for collaborative cyber defense and the requirement to preserve privacy by designing, implementing, and evaluating a practical and secure framework named FL-ZTA. The proposed framework demonstrated that by intelligently integrating FL—as a privacy-preserving core—and ZTA—as a security shield—an efficient decentralized collaboration model can be achieved. The simulation results clearly showed that the choice of aggregation algorithm plays a decisive role in the success of the system. The failure of the standard FedAvg algorithm under realistic Non-IID conditions, contrasted with the stability and robustness of FedProx, provides a practical guideline for future real-world deployments. Moreover, the quantitative analysis of DP-FedAvg empirically demonstrated the inherent trade-off between privacy assurance and model utility, offering a measurable criterion for organizations seeking to balance these two objectives. Ultimately, the FL-ZTA framework goes beyond a theoretical design—it presents a validated roadmap for the next generation of collective defense systems. This research confirms that organizations no longer need to choose between security isolation and risky collaboration; instead, FL-ZTA paves a secure path toward a resilient and intelligent digital future. The contribution of the proposed FL-ZTA framework lies primarily in the operational integration of FL and continuous zero-trust security enforcement for collaborative cybersecurity intelligence sharing across distributed organizational environments. Rather than introducing entirely new federated optimization algorithms, the study focuses on developing a trust-aware collaborative cybersecurity architecture that combines privacy preservation, secure aggregation awareness, adaptive verification

mechanisms, and distributed threat intelligence coordination within a unified framework. The presented results demonstrate the practical feasibility of integrating FL and zero-trust principles for privacy-sensitive collaborative cybersecurity applications operating under heterogeneous and security-constrained environments. Despite the encouraging experimental findings, the present study remains primarily simulation-oriented and evaluates the proposed FL-ZTA framework using a limited number of federated participants and a single cybersecurity dataset. Large-scale enterprise deployment, cross-jurisdictional trust governance, real-time federated orchestration, and adaptive multidomain policy enforcement remain important directions for future investigation. Future work may further explore hierarchical federated coordination, large-scale distributed trust management, advanced adversarial robustness mechanisms, and formal privacy budget optimization under realistic collaborative cybersecurity infrastructures.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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